

Mean, Median, and Baselines

Project: Responsible Plant Classifier

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Regression and Residuals

Without opening the old lesson, explain one key idea from Regression and Residuals. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Precision and Recall

Without opening the old lesson, explain one key idea from Precision and Recall. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Scores and Calibration Intuition

Without opening the old lesson, explain one key idea from Scores and Calibration Intuition. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Score separately for Constant prediction baselines. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Constant prediction baselines. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Apply independently

Training [1, 3, 5, 15], evaluation [3, 9]. Fit both constants and calculate MAE.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

For training targets [1,3,11], compute mean and median constant baselines. Compare total absolute training error for each. Where must final performance be measured?

6. Solve the given case

Training heights are 2,4,12 cm. Compare constant mean and median baselines.

Compute both and training absolute errors.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

For training heights 2, 3 and 10 cm, calculate mean and median. Which constant is less influenced by the high value?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 20 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Document any numeric baseline used in Plant Classifier feature checks, including unit, distribution, and outlier decision.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

For training targets [1,3,11], compute mean and median constant baselines. Compare total absolute training error for each. Where must final performance be measured?

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Fit two constants

Find the mean and median of the worked training targets.

My reasoning and evidence

13. W2 Score separately

Calculate MAE for both baselines on [4, 8].

My reasoning and evidence

14. W3 Inspect an extreme value

Replace 12 by 40. What changes?

My reasoning and evidence

15. W4 Apply independently

Training [1, 3, 5, 15], evaluation [3, 9]. Fit both constants and calculate MAE.

My reasoning and evidence

16. W5 Prevent leakage

Why not predict the evaluation mean as the baseline?

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Mean, Median, and Baselines

Fit two constant height baselines and compare them on held-out examples.

Training heights [2, 4, 6, 20] cm. Evaluation heights [4, 10] cm. Fit constants only from training.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Mean, Median, and Baselines

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Fit mean

Training mean = $(2+4+6+20)/4 = 8$ cm.



2. Fit median

Training median = $(4+6)/2 = 5$ cm.



3. Evaluate constants

Mean MAE = $(4+2)/2 = 3$ cm; median MAE = $(1+5)/2 = 3$ cm.

CHANGE ONE THING

Replace training 20 with 8 while keeping evaluation unchanged.

Explain the changed result using the matching lesson.

Does either original baseline win under evaluation MAE? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

19. Project portfolio milestone

Project: Responsible Plant Classifier

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Fit two constant height baselines and compare them on held-out examples.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Mean $MAE = (4+2)/2 = 3$ cm; median $MAE = (1+5)/2 = 3$ cm.

1. Fit mean
2. Fit median
3. Evaluate constants

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Evaluate constants Mean $MAE = (4+2)/2 = 3$ cm; median $MAE = (1+5)/2 = 3$ cm. Read the picture in order. Mean $MAE = (4+2)/2 = 3$ cm; median $MAE = (1+5)/2 = 3$ cm. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed-case practice: Replace training 20 with 8 while keeping evaluation unchanged. Which conclusion is supported by the supplied facts?

1. Both fitted constants become 5 cm and both evaluation MAEs remain 3 cm.
2. This one example guarantees correct results for every future input.
3. We can decide the result without examining the changed input or the procedure.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Both fitted constants become 5 cm and both evaluation MAEs remain 3 cm. Mean is now $20/4=5$; median stays 5. The earlier training outlier affected the mean.

Repair a missing building block

Review Regression and Residuals (ai-4/19). Goal: Distinguish numeric prediction from classification. Short practice: Which statistic reveals the size of errors without cancellation? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Precision and Recall (ai-4/17). Goal: Choose the denominator that answers the question. Short practice: Were the original precision and recall denominators the same? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Scores and Calibration Intuition (ai-4/14). Goal: Interpret calibration using groups of forecasts. Short practice: Does a calibrated 0.8 score mean every such leaf is spotted? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
